

Fundamental Theorem of Calculus (FTC)

Section 4.4

10/24/25

A) Definite Integrals

$$\int_a^b f'(x) dx = f(x) + C \Big|_a^b =$$

$$= (f(b) + C) - (f(a) + C)$$

$$= f(b) - f(a)$$

$$\int_a^b v(t) dt = s(t) \Big|_a^b = s(b) - s(a)$$

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

$$\int_a^b f''(x) dx = f'(x) \Big|_a^b = f'(b) - f'(a)$$

Example

ex1:

$$\int_1^5 3 dx = 3x \Big|_1^5 = 15 - 3 = \textcircled{12}$$

ex2:

$$\int_{-1}^4 2x dx = x^2 \Big|_{-1}^4 = 16 - 1 = \textcircled{15}$$

ex3:

$$\begin{aligned} \int_{-1}^1 (x^3 + 5x - 2) dx &= \frac{x^4}{4} + \frac{5x^2}{2} - 2x \Big|_{-1}^1 \\ &= \left(\frac{1}{4} + \frac{5}{2} - 2 \right) - \left(\frac{1}{4} + \frac{5}{2} - 2 \right) \\ &= \textcircled{-4} \end{aligned}$$

ex4:

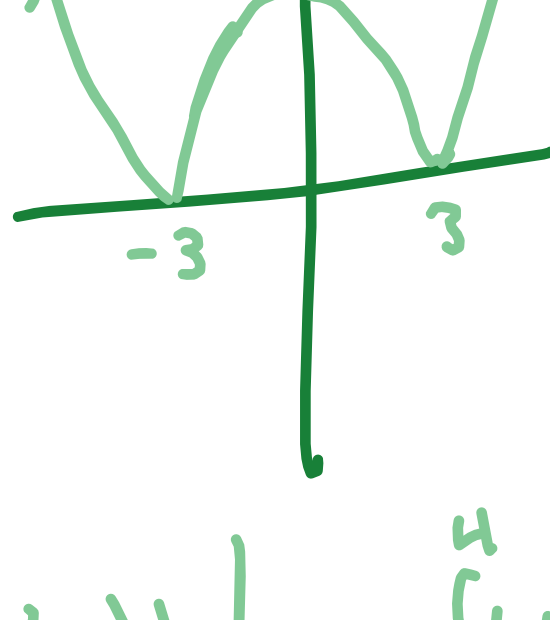
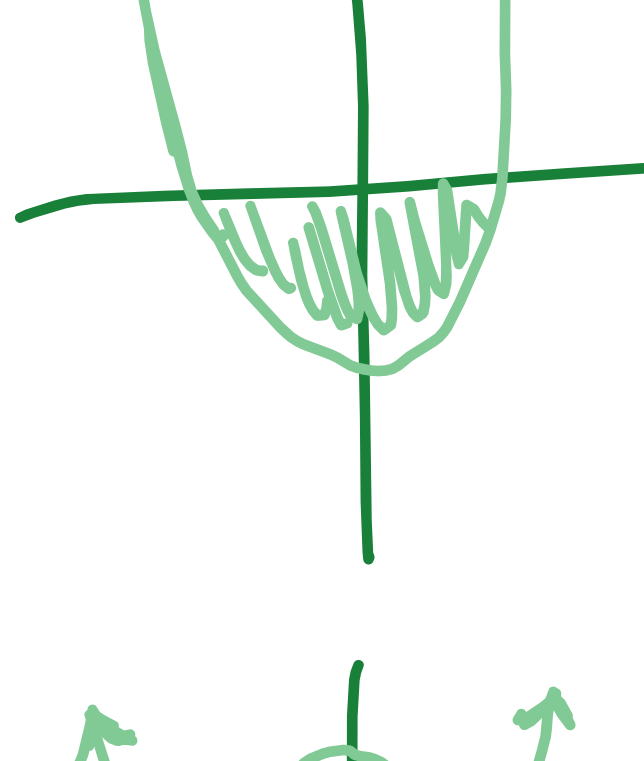
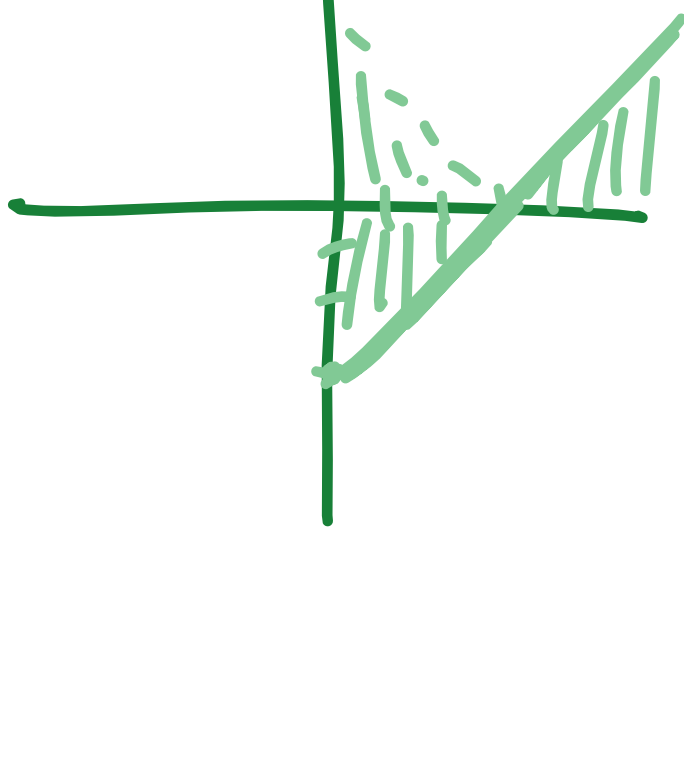
$$\int_0^{\pi} \sin x dx = -\cos x \Big|_0^{\pi} = -(-1) - (-1) = \textcircled{2}$$

ex5:

$$\begin{aligned} \int_1^3 4 + \frac{4}{x^2} dx &= 4x - \frac{4}{x} \Big|_1^3 \\ &= \left(12 - \frac{4}{3} \right) - (4 - 4) \end{aligned}$$

ex6:

$$\int_0^4 |x-3| dx = \quad \int_0^4 |x^2-9| dx =$$



$$\left| \int_0^3 (x^2-9) dx \right| + \int_3^4 (x^2-9) dx$$

(must integrate to f with)